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EXPERIMENT – 06

DELUGE SCRIPTING BASICS - LEARNING SIMULATOR

Aim

To practice basic Zoho Deluge scripting concepts through a learning simulator by working with variables, data types, operators, expressions, comments, conditional statements and list operations, and to verify the submitted scripts using the simulator's model answers and score feedback.

Procedure

1. Open the Deluge Script – Basics Learning Simulator and enter the required learner details.
2. Enter the arena and read each challenge carefully before writing the Deluge script.
3. Enter the required statement or code block in the script area and submit the answer.
4. Check the correctness message, model answer and score displayed by the simulator.
5. Proceed to the next challenge and apply the corresponding Deluge concept.
6. Record the completed challenges with screenshots and concept explanations.

Learning Overview

The simulator provides a sequence of practical exercises covering fundamental Deluge constructs. The challenges progress through list creation and modification, logical and relational operators, arithmetic expressions, comments, conditional statements and variable handling. Each completed challenge provides immediate feedback, allowing the learner to verify the script before moving to the next exercise.

Player Registration

The registration screen is used to enter the learner's roll number, name, department and email ID. These details identify the learner's practice session before the Deluge challenges begin.

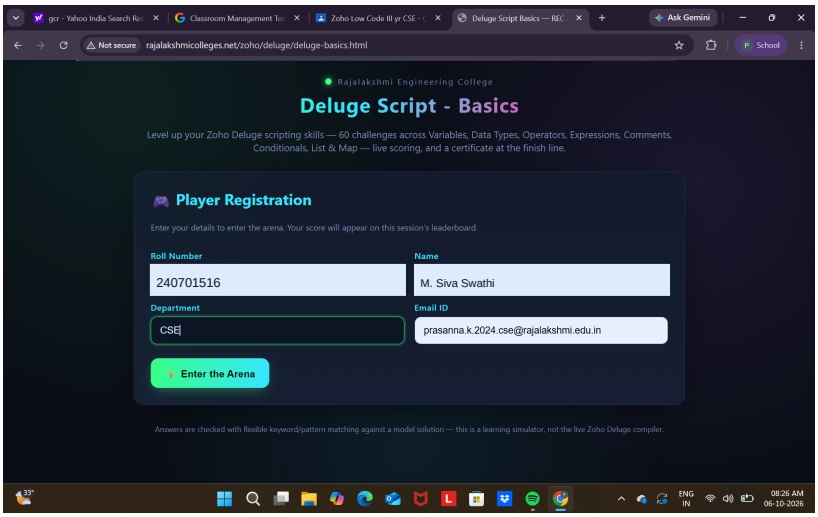


Figure: Player Registration page showing the learner details entered before starting the simulator.

Starting the Practical

After registration, the learner enters the arena and solves the Deluge scripting questions one by one. The following pages document the completed challenges and the concepts demonstrated in each script.

Challenge 1 – List

A List stores multiple values. The script creates an empty rollNumbers list and adds 101 and 102 using add().

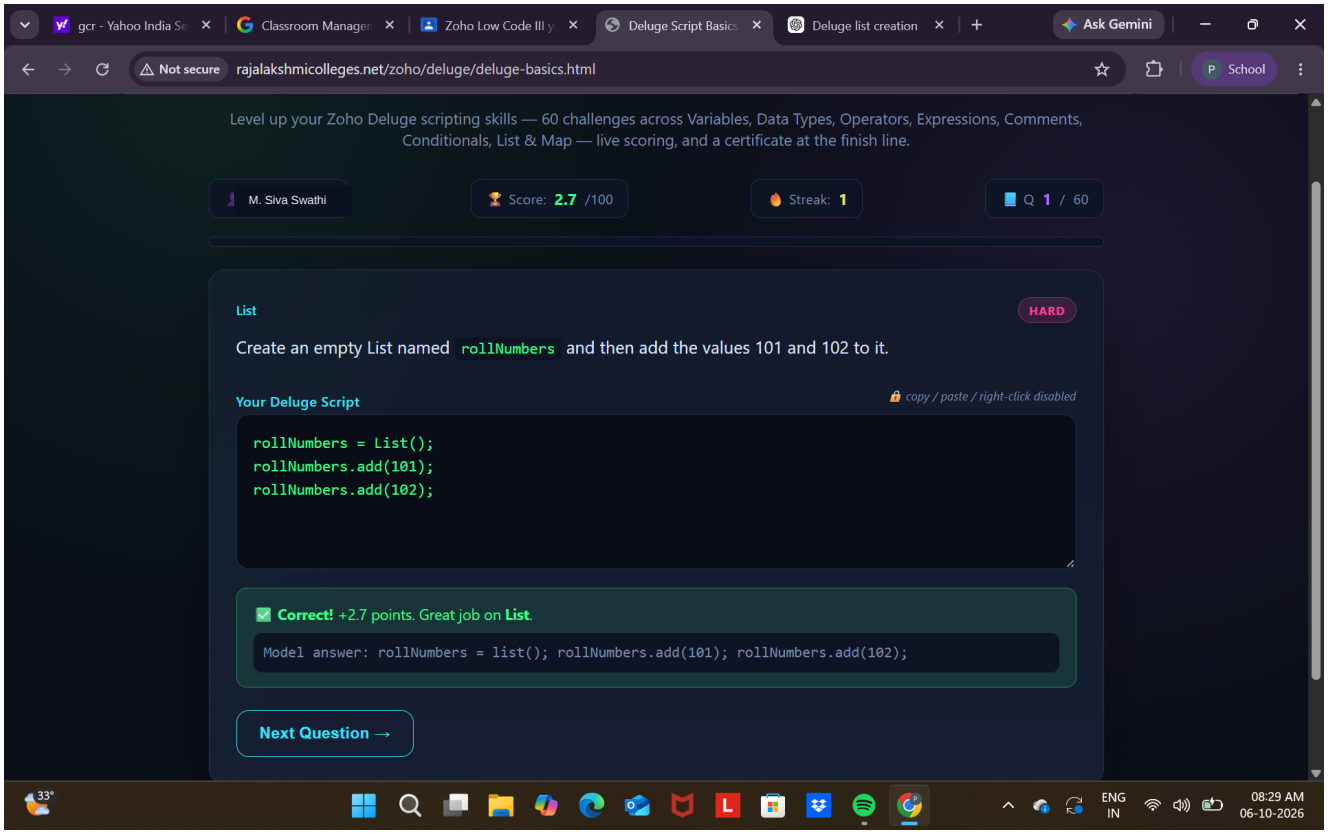


Figure: Challenge 1 – List completed in the Deluge Script Basics Learning Simulator.

Challenge 2 – Operators

The logical OR operator (||) checks whether the day is Saturday or Sunday and stores the result in isWeekend.

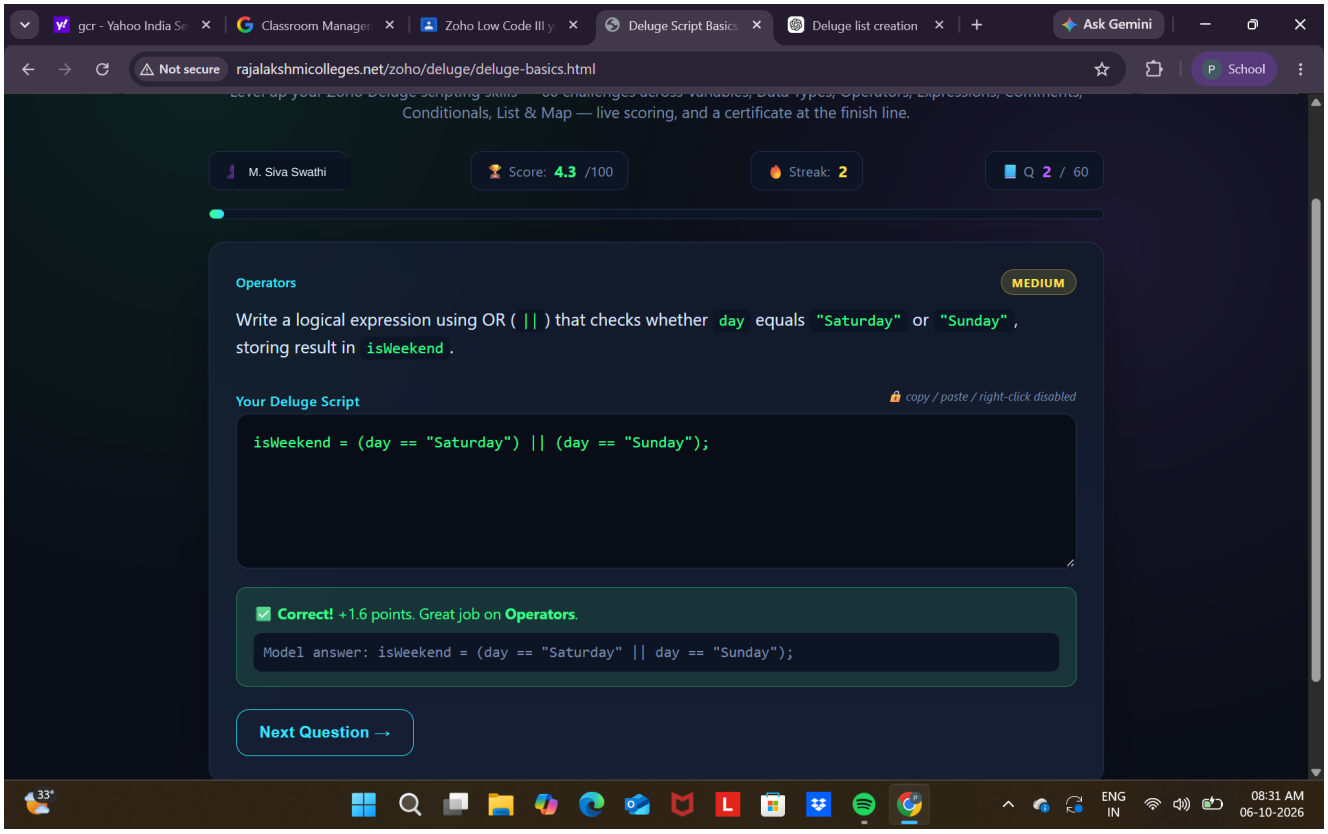


Figure: Challenge 2 – Operators completed in the Deluge Script Basics Learning Simulator.

Challenge 3 – Expressions

A relational and logical expression checks whether marks are between 40 and 100, including both limits.

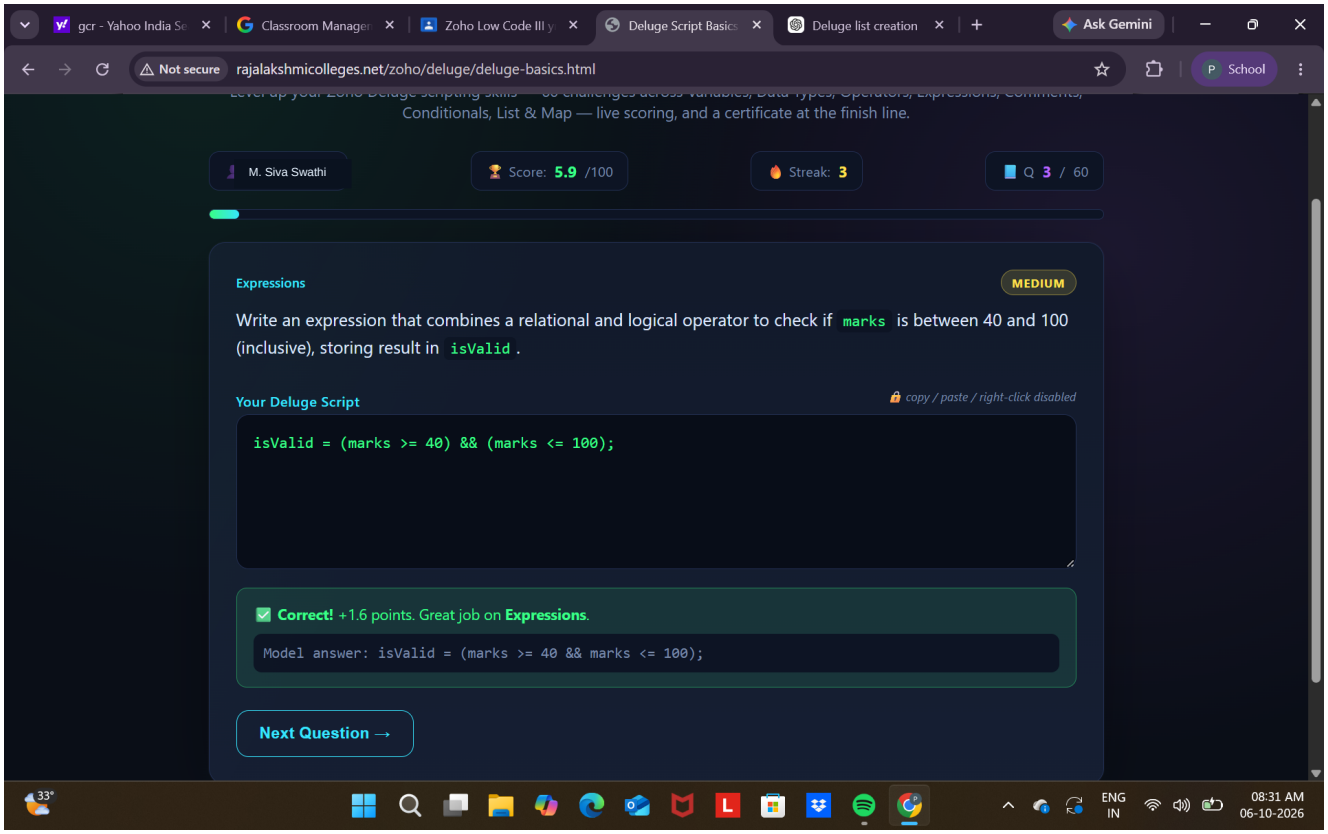


Figure: Challenge 3 – Expressions completed in the Deluge Script Basics Learning Simulator.

Challenge 4 – Comments

A multi-line comment uses /* and */. The enclosed text is ignored during execution and documents the script.

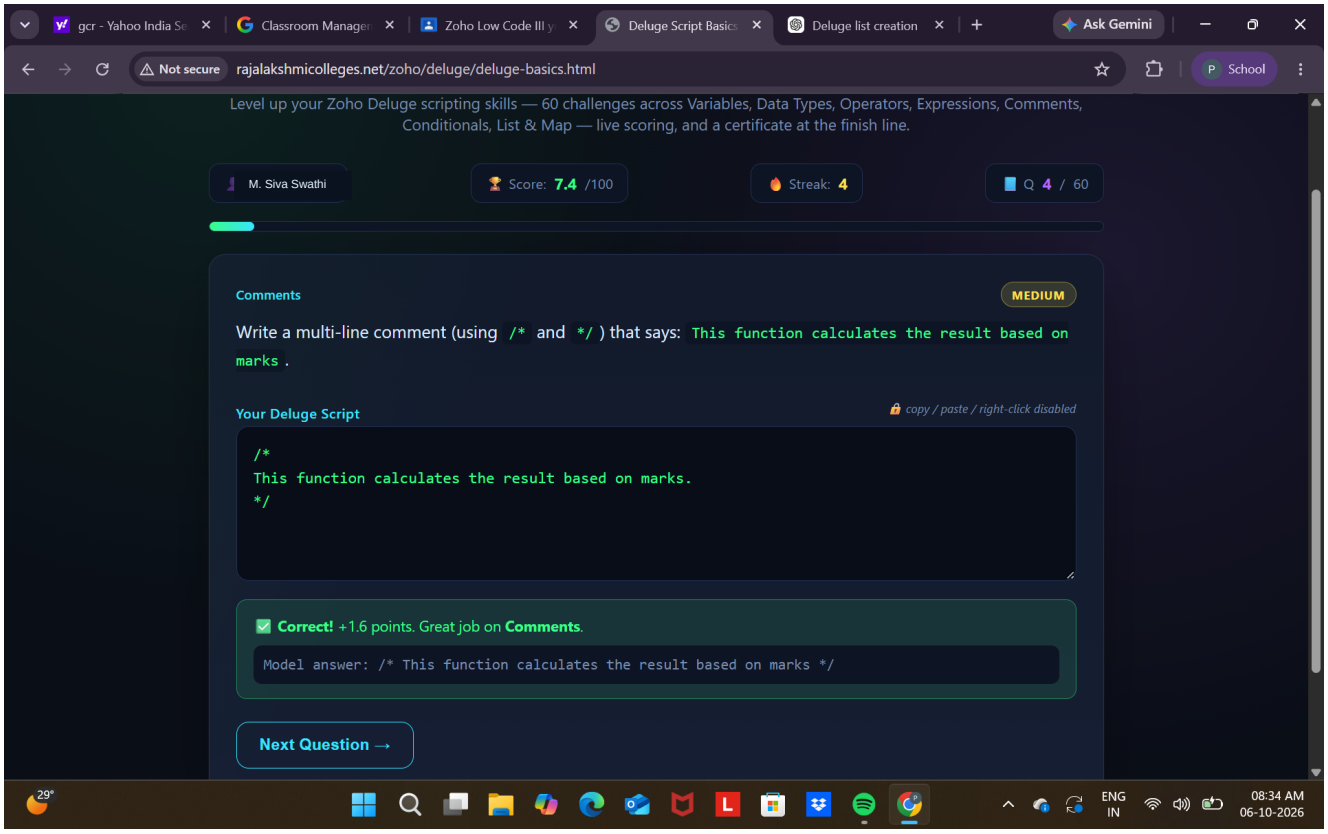


Figure: Challenge 4 – Comments completed in the Deluge Script Basics Learning Simulator.

Challenge 5 – Conditional Statements

A nested if structure checks isRegistered first and then checks feesPaid before setting status to Confirmed.

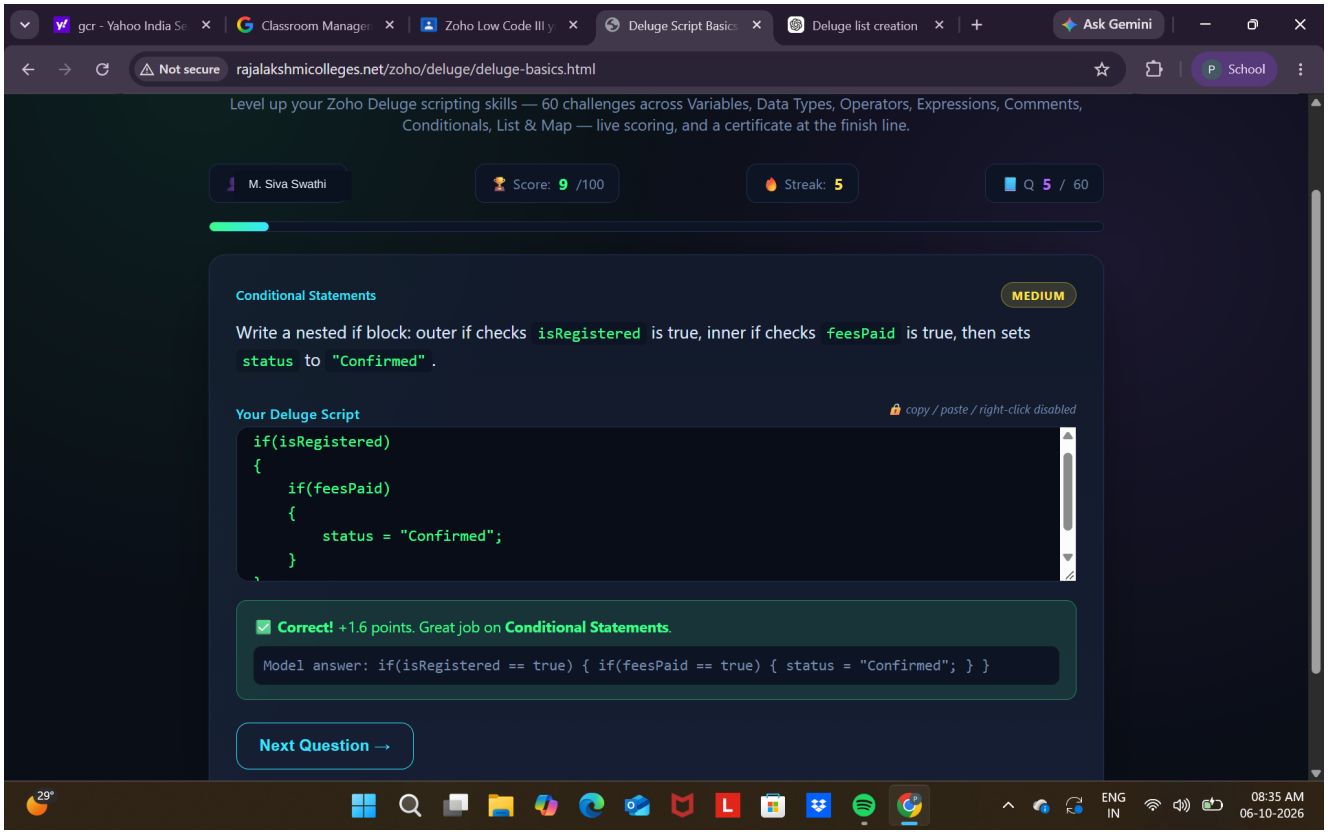


Figure: Challenge 5 – Conditional Statements completed in the Deluge Script Basics Learning Simulator.

Challenge 6 – List

The removeElement operation removes a specified value from a list. Here, Maths is removed from subjects.

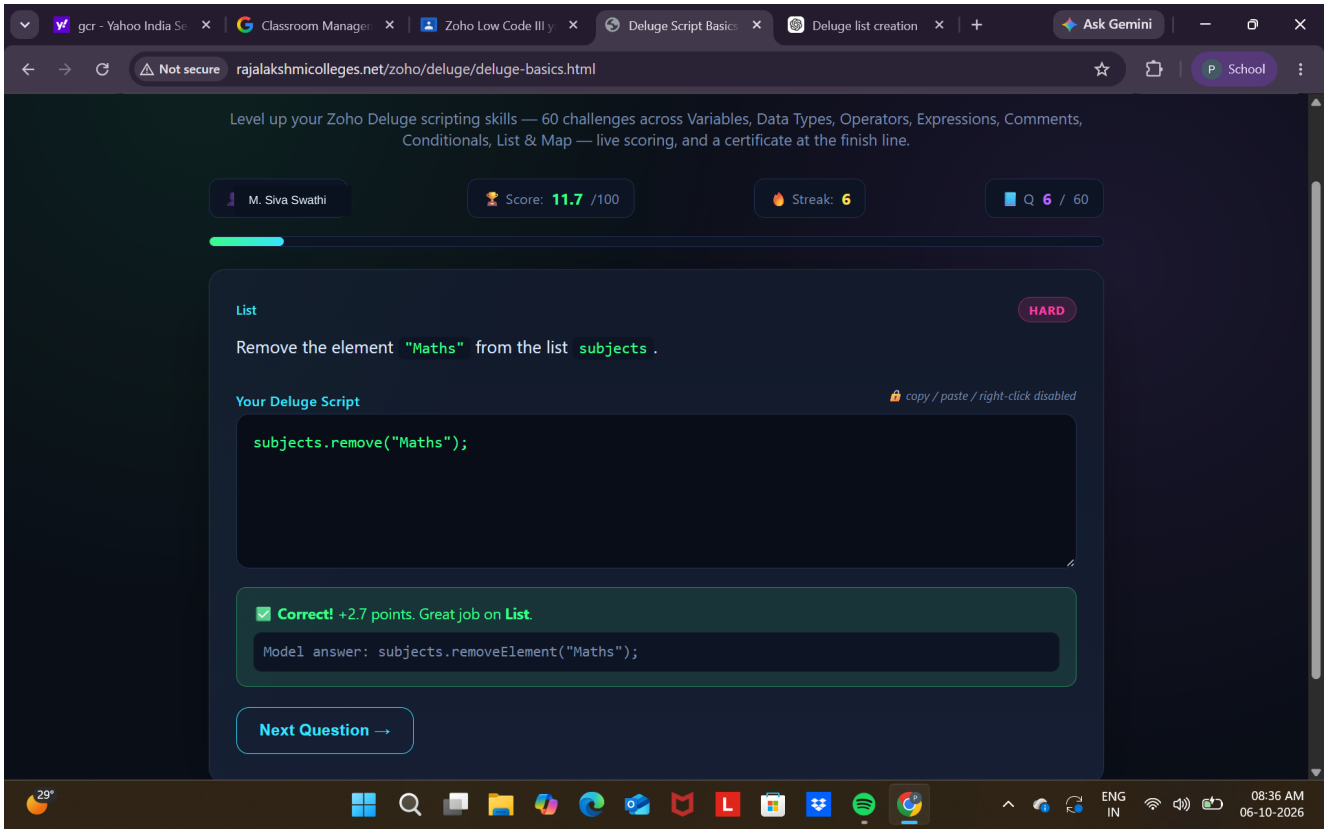


Figure: Challenge 6 – List completed in the Deluge Script Basics Learning Simulator.

Challenge 7 – Variables

Variables store values during execution. The example initializes a and b and assigns the value of b to a.

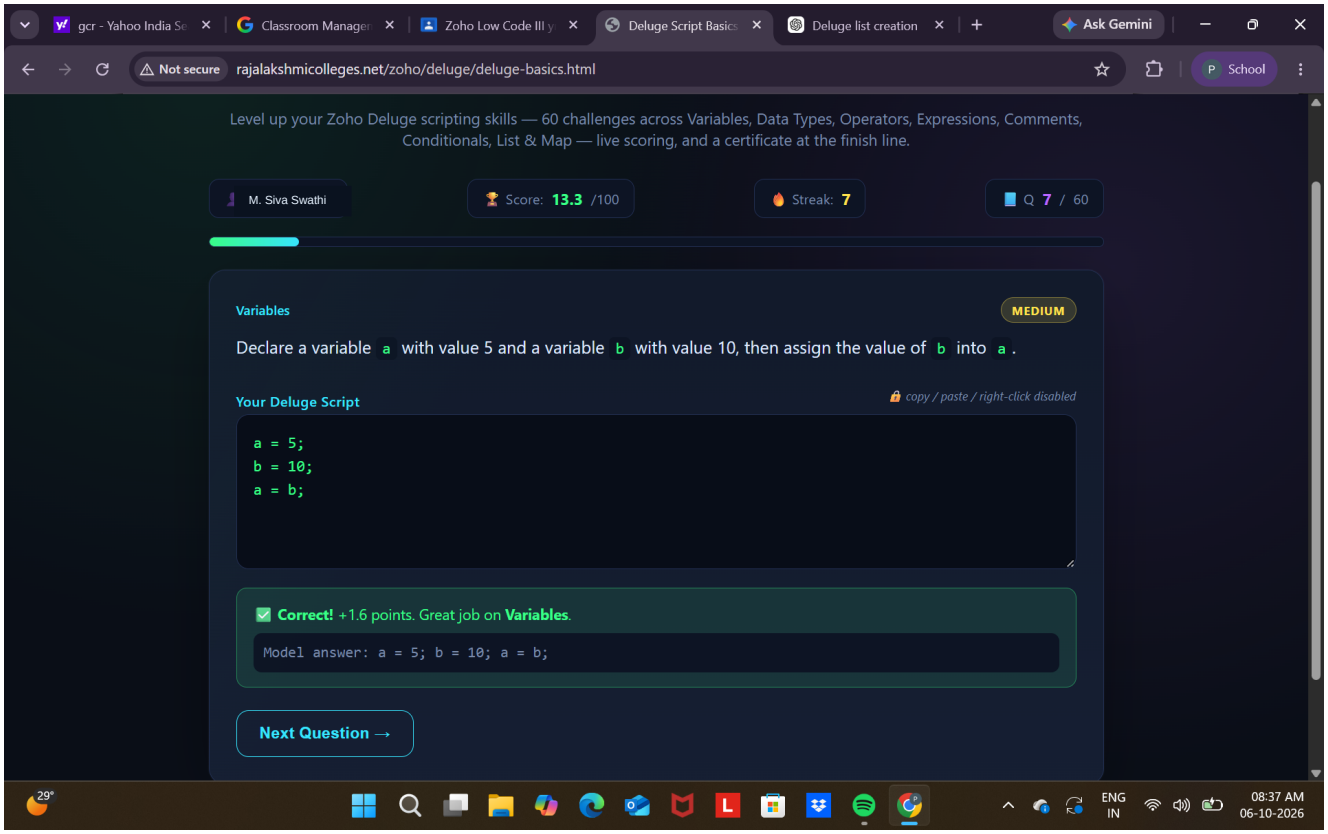


Figure: Challenge 7 – Variables completed in the Deluge Script Basics Learning Simulator.

Challenge 8 – Comments

A single-line comment begins with //. It prevents the info marks statement from executing.

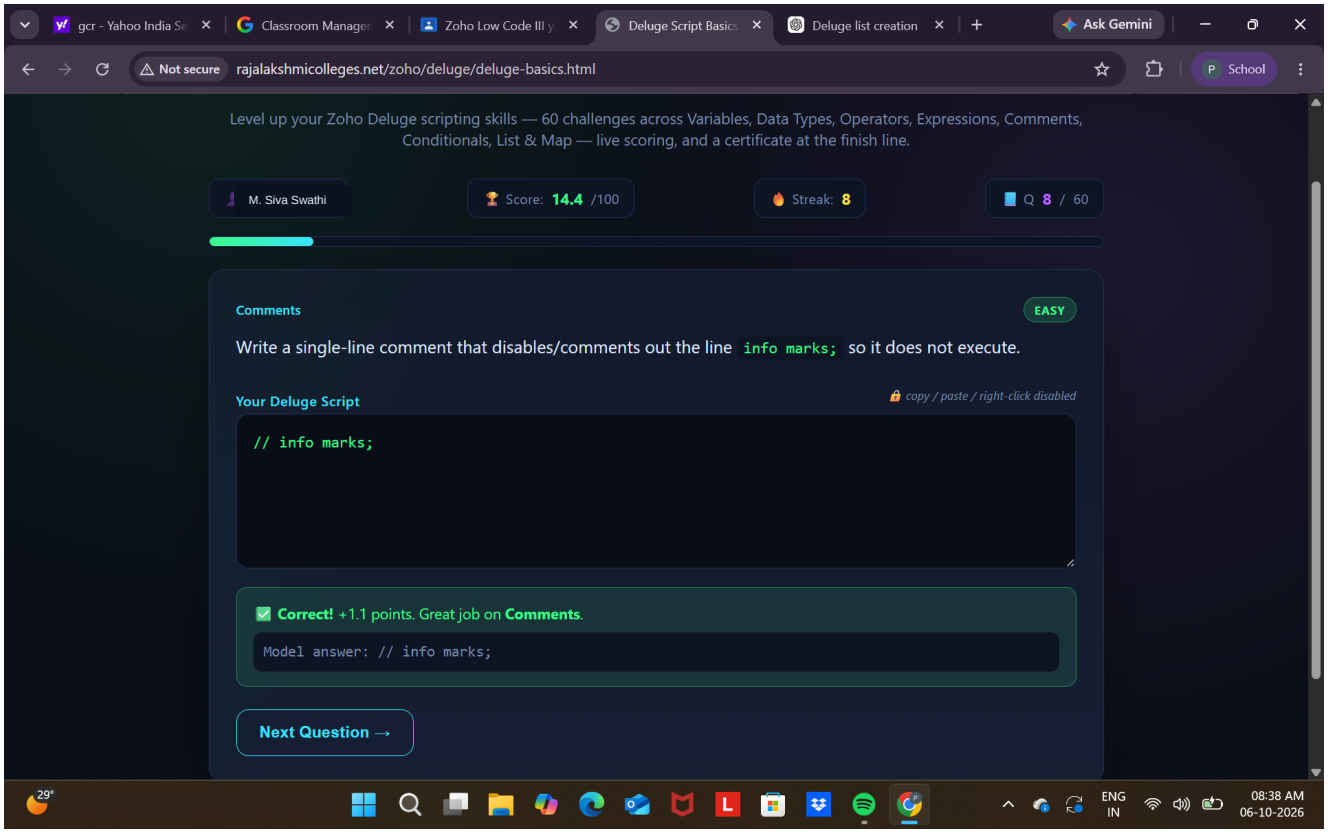


Figure: Challenge 8 – Comments completed in the Deluge Script Basics Learning Simulator.

Challenge 9 – Variables

A temporary variable is used to swap x and y, demonstrating the standard three-step variable swapping technique.

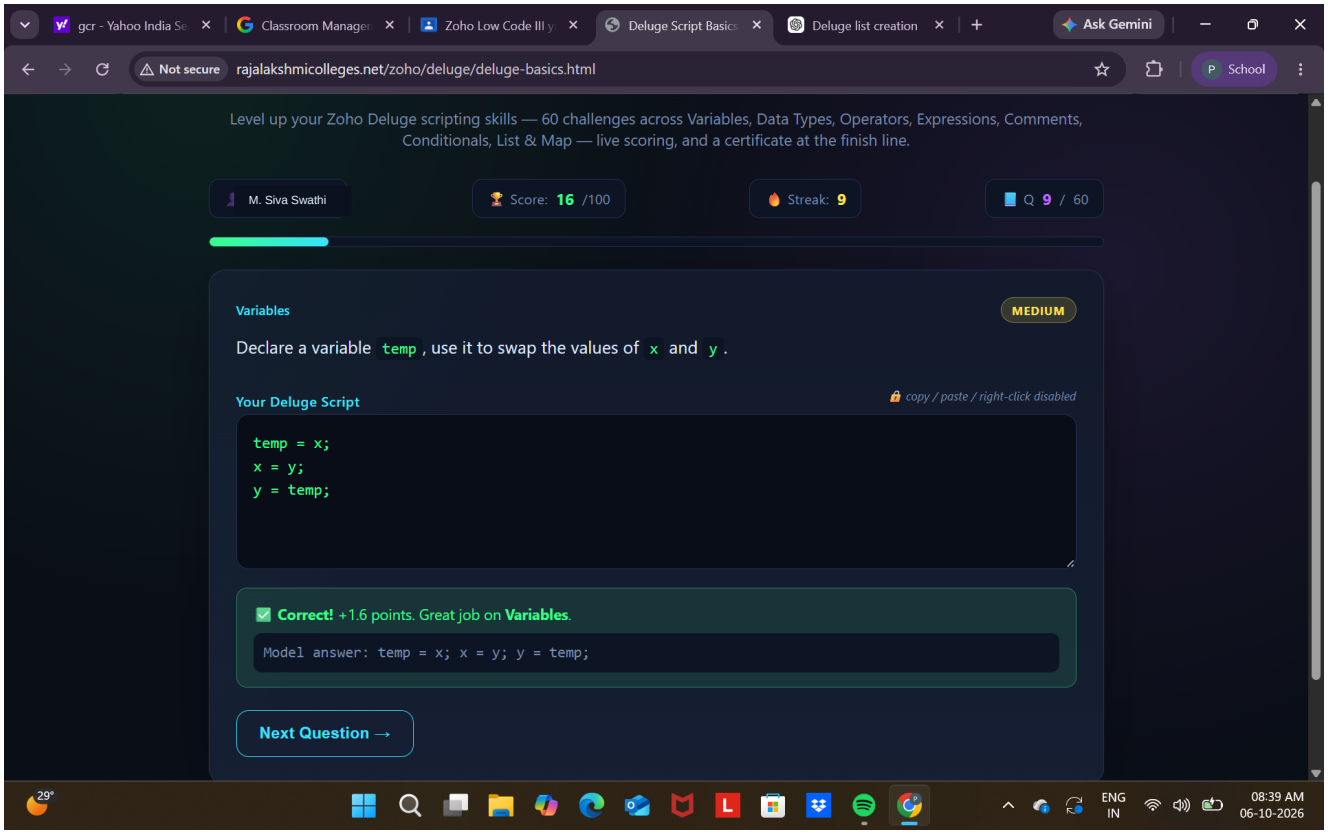


Figure: Challenge 9 – Variables completed in the Deluge Script Basics Learning Simulator.

Challenge 10 – Expressions

The percentage is calculated by dividing obtainedMarks by totalMarks and multiplying by 100.

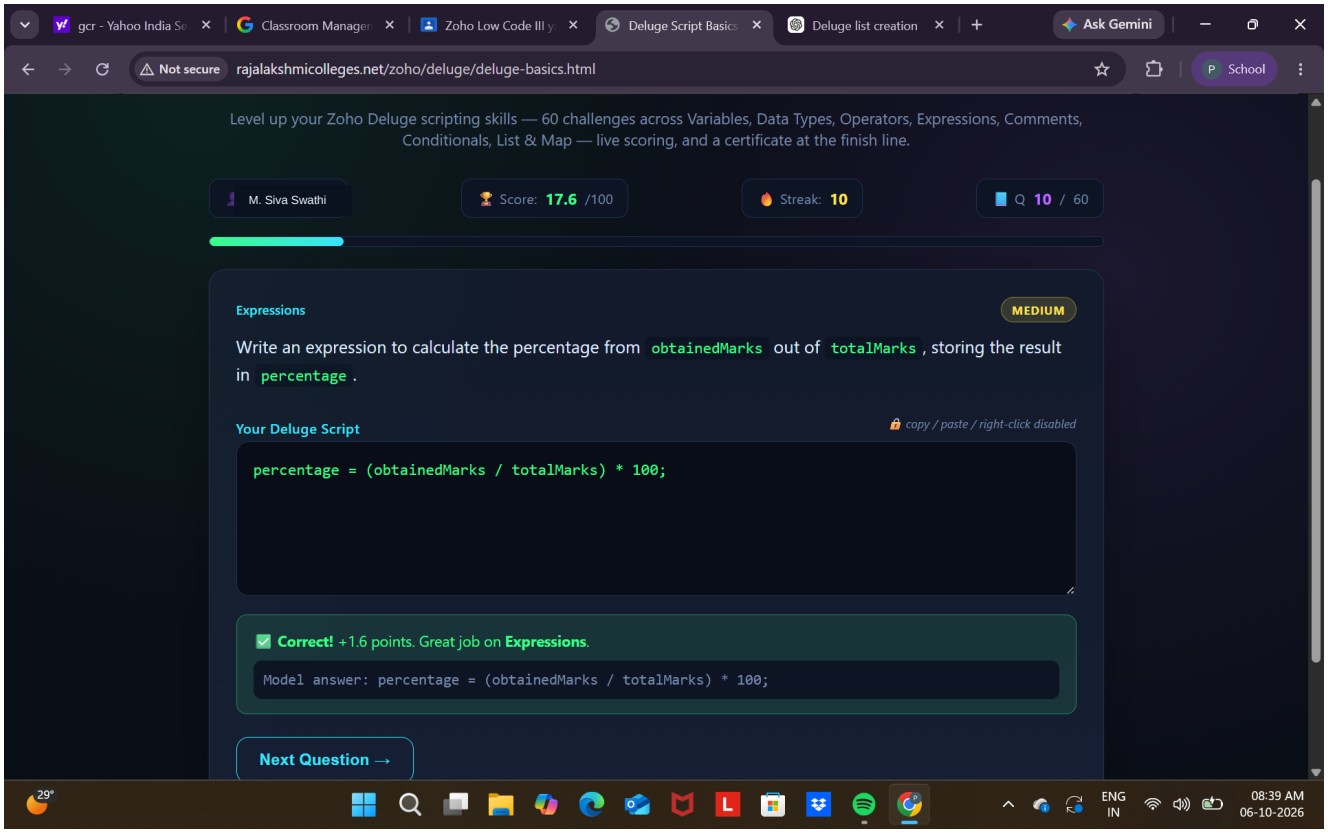


Figure: Challenge 10 – Expressions completed in the Deluge Script Basics Learning Simulator.

Challenge 11 – Operators

The logical AND operator (&&) checks that age is greater than 18 and hasID is true before storing the result in canVote.

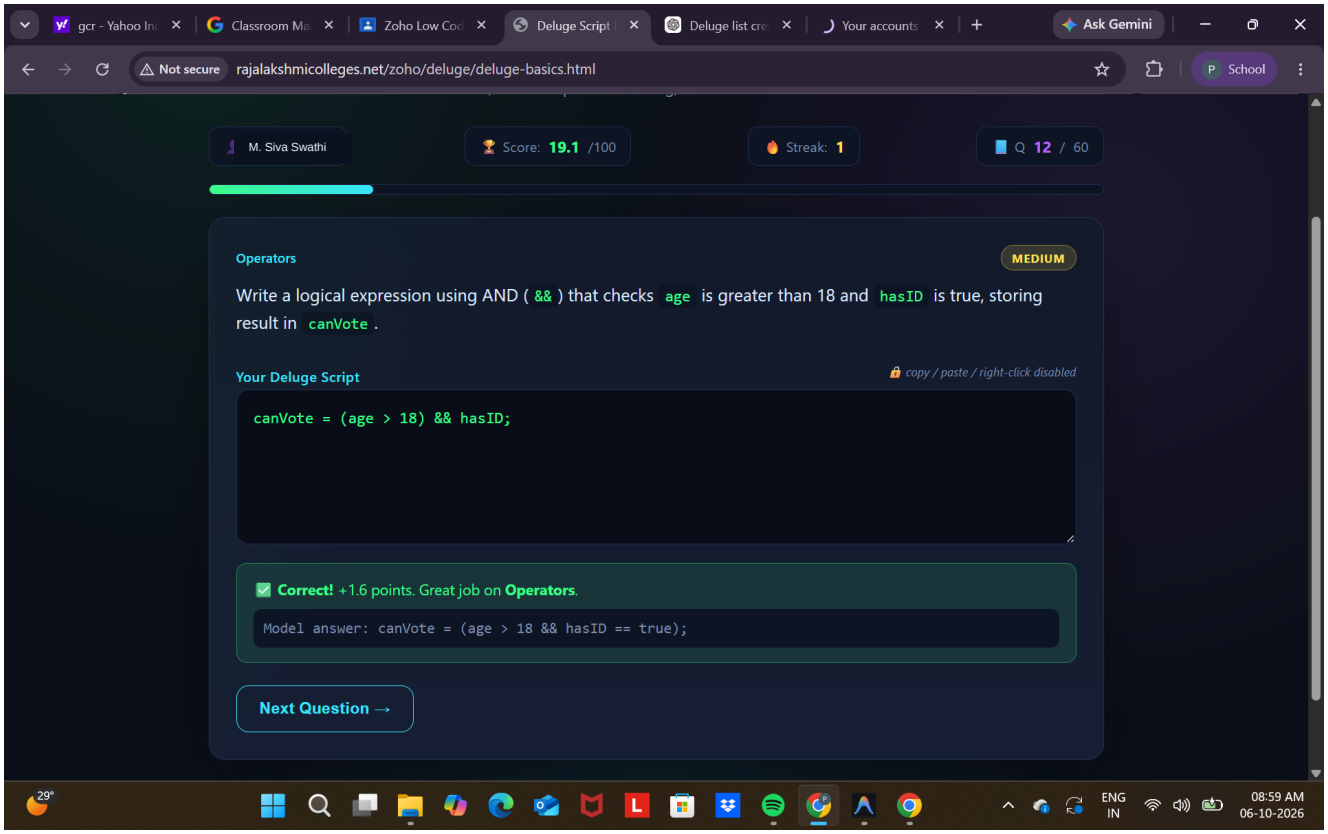


Figure: Challenge 11 – Operators completed in the Deluge Script Basics Learning Simulator.

Challenge 12 – Variables

The variable age is declared and initialized with the integer value 25.

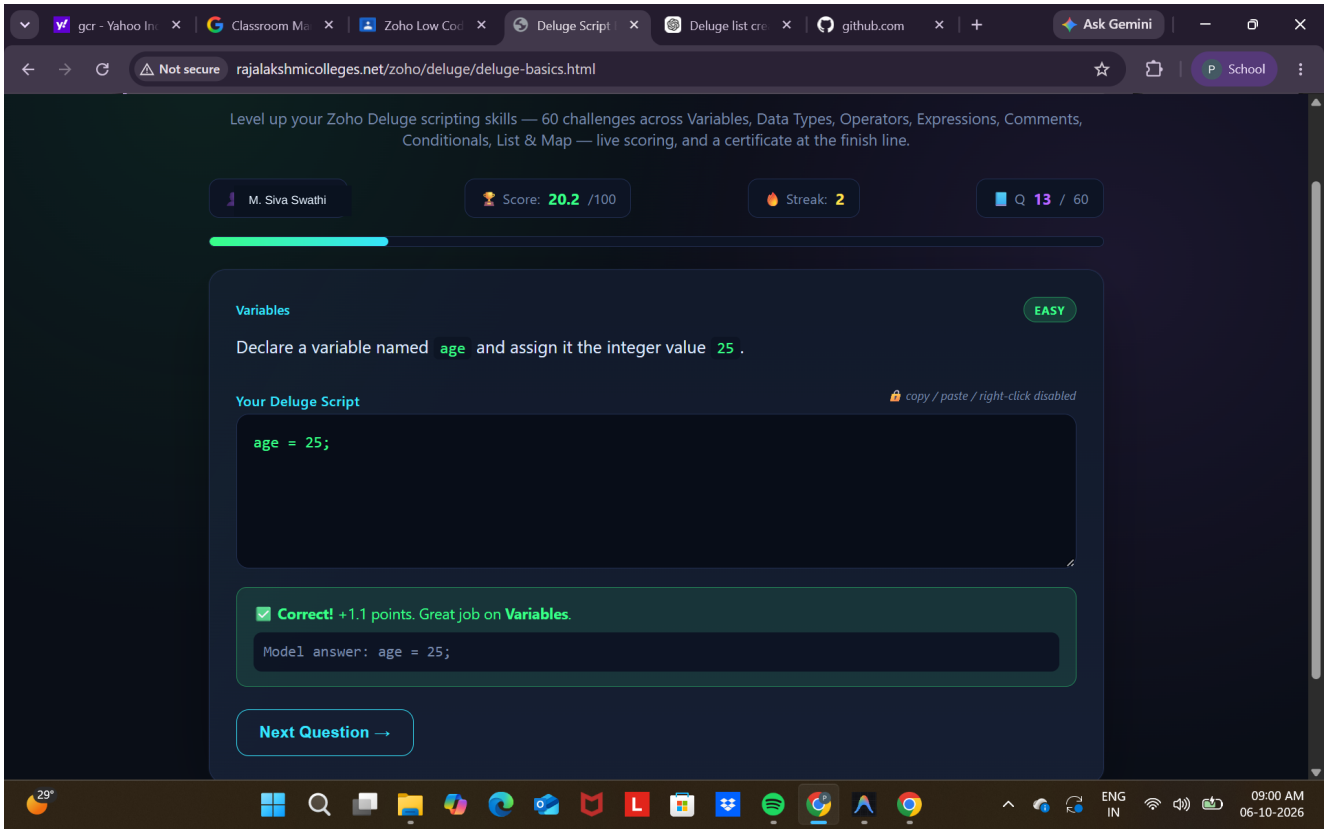


Figure: Challenge 12 – Variables completed in the Deluge Script Basics Learning Simulator.

Challenge 13 – Data Types

The Number data type is used to store the integer value 101 in the variable rollNo.

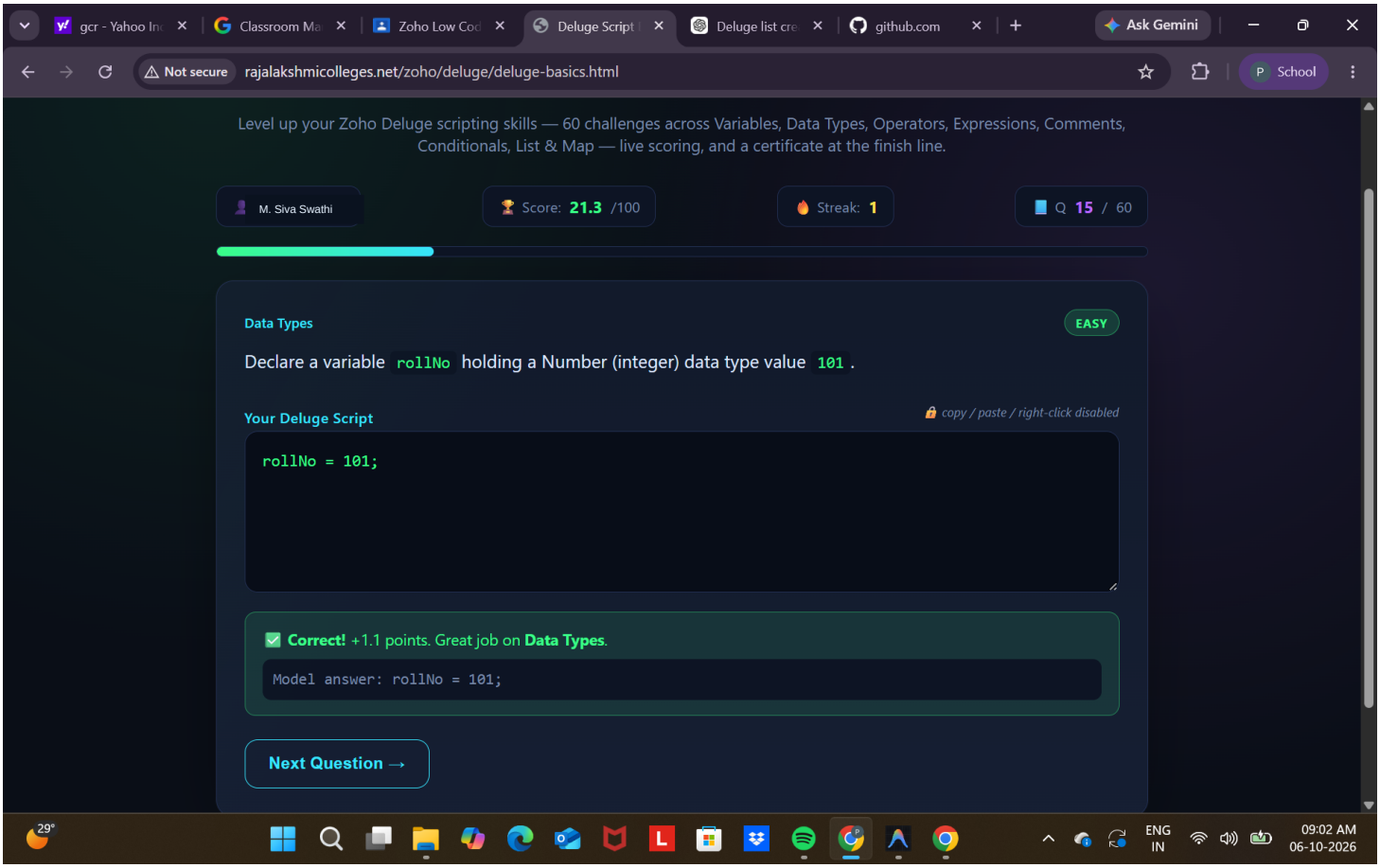


Figure: Challenge 13 – Data Types completed in the Deluge Script Basics Learning Simulator.

Result

The Deluge Scripting Basics exercises were completed successfully. The practical demonstrated variables, Number data types, logical and relational operators, arithmetic expressions, comments, nested conditional statements and list operations. The simulator verified the submitted solutions through model answers and score feedback.

Thus, the basic concepts of Deluge scripting were practiced and verified using the Learning Simulator.